

LIDAR-based MyArm Follower (LIDAR, MyArm, TFs, Inverse Kinematics)

Introduction to Robotics Lab

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Exercise Description

Objective: Implement a LIDAR-based obstacle tracking/follower system using MyArm robotic arm

Key Components:

- 1 **Elephant Robotics MyArm 300 Pi** for tracking
- 2 **Slamtec RPLIDAR A1M8** for detecting obstacles
- 3 **TF2 ROS2 package** for coordinate transformations
- 4 **Inverse Kinematics** for updating arm positioning

Functionality:

- 1 *Detect obstacles* in predefined circular sectors using LIDAR
- 2 *Transform* obstacle coords from LIDAR frame to MyArm base frame
- 3 Calculate *Inverse Kinematics* to position the end-effector properly above the detected obstacles
- 4 *Real-time visualization* of the whole system in RViz2

Radar Node

Purpose: Process LIDAR scanning data to detect the closest obstacles and publish their poses

Algorithm:

- 1 *Filter LIDAR ranges* within defined circular sector and find closest valid measurements (detected closest obstacles)
- 2 *Convert polar to cartesian* coordinates for the detected obstacle
- 3 *Publish obstacle pose* for RViz2 visualization and MyArm Node

Key Features:

- Subscribes to `/scan` topic (LaserScan messages)
- Provides configurable detection circular sector: `min_radar_angle`, `max_radar_angle`, `trigger_distance`
- Publishes obstacle pose on `/obstacle_pose` topic sending PoseStamped messages
- Visualizes detection with red sphere marker on `/radar_marker` topic sending Marker messages

MyArm Node

Purpose: Compute Inverse Kinematics and command MyArm to track/follow detected obstacles

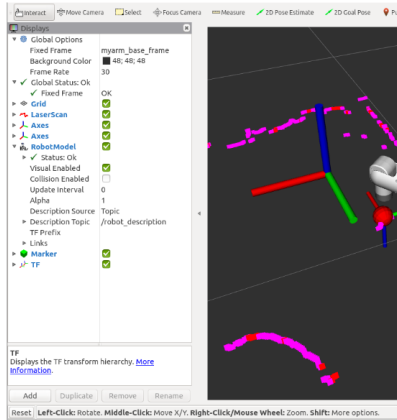
Algorithm:

- 1 *Transform obstacle pose* into MyArm base frame using TF2
- 2 Construct *desired end-effector* transformation matrix
- 3 *Solve Inverse Kinematics* with Newton-Raphson method
- 4 *Command MyArm* with the Inverse Kinematics joint solution

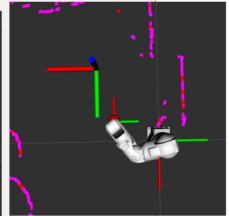
Key Features:

- Subscribes to `/obstacle_pose` topic (PoseStamped messages)
- Uses TF2 to transform obstacle pose from LIDAR frame to MyArm base (world) frame
- Publishes joint states on `/joint_states` topic sending JointState messages, for RViz2 real-time visualization
- Sends joint angle commands to the physical MyArm with adjustable speed, via configurable serial communication (port, baudrate)

Final Result



view from above



RViz2
visualization

real system

System demonstrates real-time obstacle following using MyArm robotic arm, with LIDAR scanning, coordinate transformation, and inverse kinematics control

- **Elephant Robotics Documentation**
- **Slamtec RPLIDAR A1M8 and specs**
- **Modern Robotics: Mechanics, Planning, and Control; Kevin M. Lynch and Frank C. Park**